^{34}S(p,p),(p,p′γ):resonances 1967Ko19,1977Ou01,1981Bi05

^{34}S(p,p′γ) measurements:

1967Ko19: E=2.78-3.08−MeV proton beams were produced from the 4-MeV electrostatic accelerator of the Physical-Technical Institute of the Ukrainian SSR Academy of Sciences, in energy step of 1 keV. Target was enriched thin ^{34}S. γ rays were detected with a 70 by 60 mm NaI(Tl) crystal for detecting the 2.13 MeV γ-ray of ^{34}S decay, resolution of 11.5% for the 0.622 MeV ^{137}Cs line. Measured $\sigma(E_p,\theta)$ from yields of the 2.13 MeV γ-ray, pγ(θ) of the 2.13 MeV γ-ray. Deduced resonance energies, J, π, relative resonance intensities.

^{34}S(p,p) measurement:

1977Ou01: E=1.45-2.82 MeV proton beams were produced from the TUNL 3-MV Van de Graaff accelerator in energy steps of 100 eV for on-resonance and 400 eV for off-resonance, overall FWHM=350-450 eV. Target was CdS 90% enriched ^{34}S evaporated onto 5-10 μg/cm^2 carbon backings. Scattered protons were detected with surface barrier detectors at 90°, 105°, 135° and 160°. Measured $\sigma(E_p,\theta)$. Deduced resonance energies, J, π, Γ_p from R-Matrix analysis.

1981Bi05: E=1.4-2.8 MeV proton beams were produced from the Groningen 5 MV Van de Graaff accelerator. Targets were 4 and 1 μg/cm^2 Sb_2S_3 (90% in ^{34}S) evaporated onto 10 μg/cm^2 carbon backings. Scattered protons were detected with three silicon surface barrier detectors at 88°, 124° and 174°. Measured $\sigma(E_p,\theta)$. Deduced resonance energies, J, π, Γ_p. Also measured (p,γ).

1971BrXT: Measured $\sigma(E_p)$. Deduced resonance energies, J, π, widths.

1974Bi16: E=1.2-2.1 MeV protons were produced from the Helsinki University 2.5 MV van de Graaff accelerator. Targets were prepared from natural sulphur (4.22% ^{34}S) with ^{34}S ions embedded into carbon foils. Scattered protons were detected with four silicon surface barrier detectors. Measured $\sigma(E_p,\theta)$. Deduced resonance energies, J, π, widths from analysis of the resonances.

Others: 1978Ce02, 1980Fa07, 1983Ch50, 1988Za04, 2019Ka53.

^{35}Cl Levels

A_2 and A_4 values under comments are from pγ(θ) of the 2.13 MeV γ-ray from ^{34}S (1967Ko19).
I(res) under comments are resonance intensities relative to 1 for E_p(lab)=2799 resonance (1967Ko19).

E(level)[†]	J^{π}[‡]	Γ_p[#]	E_p(lab)(keV)[#]	Comments
7195[@]	1/2$^-$[@]		848	Γ=27 eV *8* (1971BrXT).
7274[@]	1/2$^-$[@]		929	Γ=14 eV *4* (1971BrXT).
7550[&]	7/2$^-$[&]	0.0072[&] keV *15*	1214	Γ=11.0 eV *15*, Γ_γ=3.8 eV *15* (1974Bi16).
7686[a]	3/2$^-$[a]		1354	Γ_p=445 eV *11* for J^π=3/2$^-$ (1981Bi05).
7706[a]	3/2$^+$,5/2$^+$[a]		1375	Γ_p=4 eV *1* for J^π=5/2$^+$, 17 eV *3* for J^π=3/2$^+$ (1981Bi05).
7796 *3*	(1/2)$^-$	0.031 keV *10*	1467.3 *5*	
7837 *3*	3/2$^-$	3.7 keV *4*	1510.0 *5*	Γ=3.8 keV *2*, Γ_p=3.8 keV *2* (1974Bi16).
7880 *3*	3/2$^+$,5/2$^+$	0.008 keV *4*	1554.0 *5*	
8004 *3*	3/2$^+$,5/2$^+$	0.011 keV *5*	1682.1 *5*	Γ_p=21 keV *3* for J^π=5/2$^+$, 24 eV *3* for J^π=3/2$^+$ (1981Bi05).
8034 *3*	1/2$^-$,3/2$^-$	0.026 keV *7*	1712.9 *5*	
8038 *3*	1/2$^+$	0.30 keV *2*	1716.9 *5*	
8148 *3*	1/2$^-$	2.66 keV *27*	1830.2 *5*	
8150 *3*	3/2$^-$	0.56 keV *6*	1831.5 *5*	
8209 *3*	(5/2)$^+$	0.033 keV *10*	1893.2 *5*	Γ=100 eV *20*, Γ_p=120 eV *20* (1974Bi16).
8211 *3*	1/2$^+$	0.094 keV *15*	1894.4 *5*	Γ_p=39 eV *5* for J^π=5/2$^+$, 44 eV *6* for J^π=3/2$^+$, 120 eV *11* for J^π=1/2$^+$ (1981Bi05). Γ=270 eV *50*, Γ_p=270 eV *50* (1974Bi16).
8216[&]	5/2$^+$[&]	0.014[&] keV *3*	1900	Γ=14 eV *4*, Γ_γ=0.78 eV (1974Bi16).
8243 *3*	3/2$^-$	0.140 keV *15*	1928.2 *5*	
8271 *3*	3/2$^+$,5/2$^+$	0.005 keV *3*	1956.1 *5*	
8279 *3*	3/2$^+$,5/2$^+$	0.006 keV *3*	1964.6 *5*	
8290 *3*	(1/2)$^-$	0.04 keV *1*	1975.7 *5*	Γ_p=49 eV *11* for J^π=3/2$^-$, 48 eV *6* for J^π=1/2$^-$ (1981Bi05).
8300 *3*	3/2$^-$	0.073 keV *15*	1986.7 *5*	
8383 *3*	3/2$^+$,5/2$^+$	0.023 keV *7*	2072.2 *5*	Γ=30 eV *7*, Γ_p=28 eV *6*, Γ_γ=2.4 eV (1974Bi16). J^π: 5/2$^+$ in 1974Bi16.
8404 *3*	5/2$^-$,7/2$^-$	0.002 keV *1*	2093.9 *5*	
8406 *3*	(5/2$^-$,7/2$^-$)	0.001 keV *1*	2095.2 *5*	

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$^{34}S(p,p),(p,p'\gamma)$:resonances 1967Ko19,1977Ou01,1981Bi05 (continued)

^{35}Cl Levels (continued)

E(level)[†]	J^{π}[‡]	Γ_p[#]	E_p(lab)(keV)[#]	Comments
8409 *3*	1/2$^-$	0.125 keV *15*	2098.7 *5*	
8418 *3*	1/2$^+$	0.026 keV *7*	2107.5 *5*	
8435 *3*	(3/2)$^+$	0.090 keV *15*	2125.9 *5*	
8486 *3*	3/2$^+$,5/2$^+$	0.012 keV *5*	2178.2 *5*	
8488 *3*	3/2$^-$	0.150 keV *15*	2179.7 *5*	
8515 *3*	1/2$^-$	0.150 keV *15*	2208.1 *5*	
8573 *3*	(5/2)$^+$	0.08 keV *1*	2267.9 *5*	
8582 *3*	1/2$^+$	0.75 keV *8*	2276.7 *5*	
8592 *3*	3/2$^+$,5/2$^+$	0.003 keV *2*	2287.4 *5*	
8616 *3*	5/2$^+$	0.175 keV *20*	2311.6 *5*	
8620 *3*	3/2$^+$,5/2$^+$	0.002 keV *1*	2316.2 *5*	
8633 *3*	(3/2,5/2$^+$)	0.001 keV *1*	2328.8 *5*	
8643 *3*	3/2$^+$,5/2$^+$	0.003 keV *2*	2339.9 *5*	
8687 *3*	5/2$^-$,7/2$^-$	0.001 keV *1*	2385.0 *5*	
8689 *3*	1/2$^+$	0.20 keV *2*	2387.2 *5*	
8691 *3*	1/2$^-$	6.4 keV *7*	2388.6 *5*	
8698 *3*	3/2$^-$	0.8 keV *1*	2396.0 *5*	
8750 *3*	3/2$^-$	0.30 keV *3*	2449.5 *5*	
8775 *3*	1/2$^-$	0.57 keV *6*	2475.2 *5*	
8781 *3*	3/2$^-$	0.214 keV *25*	2481.7 *5*	
8788 *3*	(3/2$^+$,5/2$^+$)	0.001 keV *1*	2489.1 *5*	
8799 *3*	(3/2$^+$,5/2$^+$)	0.001 keV *1*	2500.3 *5*	
8825 *3*	1/2$^+$	1.70 keV *17*	2526.8 *5*	
8829 *3*	1/2$^-$	12.2 keV *12*	2530.8 *5*	
8830 *3*	1/2$^+$	0.080 keV *15*	2532.0 *5*	
8839 *3*	5/2$^-$,7/2$^-$	0.001 keV *1*	2541.7 *5*	Γ_p=4 eV *1* for J^{π}=7/2$^-$, 2 eV *1* for J^{π}=5/2$^-$ (1981Bi05).
8858 *3*	3/2$^+$,5/2$^+$	0.010 keV *5*	2560.8 *5*	
8870 *3*	3/2$^+$,5/2$^+$	0.027 keV *10*	2573.2 *5*	
8909 *3*	3/2$^+$,5/2$^+$	0.002 keV *1*	2613.2 *5*	
8955 *3*	(3/2)$^+$	0.075 keV *15*	2661.3 *5*	
8959 *3*	(1/2$^-$,3/2$^-$)	0.04 keV *1*	2665.4 *5*	
8983 *3*	5/2$^-$,7/2$^-$	0.003 keV *2*	2690.0 *5*	
8985 *3*	3/2$^+$,5/2$^+$	0.025 keV *10*	2692.1 *5*	
8998 *3*	5/2$^-$,7/2$^-$	0.002 keV *1*	2705.1 *5*	
9021 *3*	3/2$^-$	3.50 keV *35*	2728.4 *5*	
9031 *3*	(5/2)$^+$	0.04 keV *1*	2739.4 *5*	
9040 *3*	1/2$^-$	0.292 keV *30*	2748.2 *5*	
9048 *3*	5/2$^-$,7/2$^-$	0.001 keV *1*	2756.7 *5*	
9050 *3*	(5/2)$^+$	0.095 keV *15*	2758.8 *5*	
9084 *3*	(5/2)$^+$	0.057 keV *10*	2793.2 *5*	Γ_p=59 eV *7* for J^{π}=5/2$^+$, 67 eV *8* for J^{π}=3/2$^+$ (1981Bi05).
9089 *4*			2799 *4*	I(res)=1. E_p(lab)(keV): from 1967Ko19.
9101 *3*	3/2$^-$	0.20 keV *2*	2810.9 *5*	
9108 *4*			2818 *4*	I(res)=0.18.
9114 *4*			2825 *4*	I(res)=0.34.
9120 *4*			2831 *4*	I(res)=0.31.
9127 *4*			2838 *4*	I(res)=0.16.
9136 *4*			2847 *4*	I(res)=0.34.
9147 *4*			2859 *4*	I(res)=0.27.
9159 *4*			2871 *4*	I(res)=0.29.
9168 *4*	(3/2$^-$)		2880 *4*	I(res)=3.70. A_2=+0.19 *133*, A_4=+0.017 *1480* (1967Ko19).
9185 *4*			2898 *4*	I(res)=0.11.
9196 *4*			2909 *4*	
9207 *4*	9/2$^+$		2920 *4*	I(res)=1.70. A_2=+0.09 *18*, A_4=+0.37 *12* (1967Ko19).

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$^{34}S(p,p),(p,p'\gamma)$:resonances 1967Ko19,1977Ou01,1981Bi05 (continued)

^{35}Cl Levels (continued)

E(level)†	J^{π}‡	E_p(lab)(keV)#	Comments
9222 *4*		2936 *4*	I(res)=0.25.
9227 *4*	$5/2^-$	2941 *4*	I(res)=0.59. A_2=+0.59 *28*, A_4=+0.29 *32* (1967Ko19).
9241 *4*	$7/2^+$	2955 *4*	I(res)=2.32. A_2=+0.24 *40*, A_4=+0.27 *53* (1967Ko19).
9255 *4*		2970 *4*	I(res)=0.45.
9271 *4*		2986 *4*	I(res)=0.34.
9276 *4*		2991 *4*	I(res)=0.32.
9284 *4*	$5/2^-$	3000 *4*	I(res)=3.86. A_2=+0.72 *28*, A_4=−0.17 *29* (1967Ko19).
9294 *4*		3010 *4*	I(res)=0.68.
9299 *4*		3015 *4*	I(res)=0.45.
9325 *4*	$5/2^+$	3042 *4*	I(res)=1.81. A_2=+0.81 *28*, A_4=−0.48 *35* (1967Ko19).
9336 *4*	$7/2^+$	3053 *4*	I(res)=1.00. A_2=+0.17 *29*, A_4=+0.34 *40* (1967Ko19).
9345 *4*		3062 *4*	I(res)=0.50.
9349 *4*		3067 *4*	I(res)=0.50.
9355 *4*		3073 *4*	I(res)=2.23.

† From E(level)=E_p(cm)+S(p)(^{35}Cl), where S(p)=6370.81 *4* (2021Wa16) and E_p(cm) deduced from E_p(lab) by $E_p(cm)=E_p(lab)*[m(^{34}S)/(m(^{34}S)+m(p))]$.

‡ From comparison of experimental differential cross sections with theoretical predictions by R-Matrix calculations (1977Ou01) for levels up to 9101 level and from $p\gamma(\theta)$ (1967Ko19) for levels after that.

From 1977Ou01 up to 9101 level and from 1967Ko19 after that, unless otherwise noted. 1977Ou01 state that although the absolute proton energies are accurate only to about 3 keV, the relative values of E_p(lab) should be reliable to within a few hundred eV, based on which the evaluators have assigned a 0.5 keV uncertainty for E_p(lab) and 3 keV uncertainty for E(level) from 1977Ou01.

@ From 1971BrXT.

& From 1974Bi16.

[a] From 1981Bi05.